Kaiwei Tu

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Summary

Research engineer specializing in large-scale AI infrastructure for frontier agentic systems. Work spans hardware/software co-design, performance characterization, and pathfinding next-generation memory/storage systems for long-context agentic workloads at hyperscale.

Education

University of Wisconsin–Madison **Madison, WI**
Ph.D. in Computer Sciences 2021–2026
M.S. in Computer Sciences 2019–2021
Advised by Prof. Remzi H. Arpaci-Dusseau and Prof. Andrea C. Arpaci-Dusseau

University of Michigan **Ann Arbor, MI**
B.S. in Engineering, Computer Science, summa cum laude *2015–2019*

Shanghai Jiao Tong University **Shanghai, China**
B.S. in Electrical and Computer Engineering *2014–2019*

Professional Experience

Meta **Menlo Park, CA**
Research Scientist, AI and Systems Co-design *Jul 2026–Present*

- Build and optimize agentic infrastructure for personal intelligence, powering Meta's **Muse** models, spanning software/hardware co-design and performance optimization.
- Pathfind emerging memory technologies (e.g., high-bandwidth flash) to scale long-context inference capacity and cost-efficiency.
- Develop agentic systems that automate workload cloning for software/hardware co-design and pre-silicon hardware design.

Meta **Menlo Park, CA**
AI and Systems Co-design, PhD Intern *May 2025–Aug 2025*

- Characterized hyperscale ads inference workloads and developed AI inference benchmarks contributed to DCPerf, Meta's open-source benchmark suite reflecting real-world hyperscale data center applications.

Google **Sunnyvale, CA**
Google Cloud, Research Intern *May 2022–Aug 2022*

- Evaluate memory bandwidth isolation mechanisms on AMD processors across Google's production fleet; proposed a hybrid CPU-jailing and resctrl policy to improve regulation efficiency.

Citrix **Fort Lauderdale, FL**
Software Development Engineer Intern *May 2018–Aug 2018*

- Enhanced cloud VM provisioning service; led development of a RESTful API and UI for deployment monitoring.

Research Experience

Advanced Systems Laboratory (ADSL), University of Wisconsin–Madison **Madison, WI**
Research Assistant *May 2021–May 2026*

- Investigated load balancing mechanisms for tiered memory systems to reduce inter-job interference.
- **MOST (FAST '26):** Developed a mirror optimized tiering scheme for heterogeneous storage hierarchies.
- **NyxCache (FAST '22):** Designed and implemented the Nyx controller for multi-tenant persistent memory caching framework; evaluated QoS, proportional sharing, and fair slowdown policies.
- **Orthus (FAST '21):** Implemented Non Hierarchy Caching (NHC) in Intel's Open CAS Linux kernel module and evaluated performance under dynamic workloads.

Publications

[1]: **Kaiwei Tu**, Andrea C. Arpaci-Dusseau, and Remzi H. Arpaci-Dusseau. A novel load balancing mechanism for tiered memory (title withheld for anonymity). *In submission*.

[2]: **Kaiwei Tu**, Kan Wu, Andrea C. Arpaci-Dusseau, and Remzi H. Arpaci-Dusseau. "Getting the MOST out of your Storage Hierarchy with Mirror-Optimized Storage Tiering." *USENIX FAST '26*.

[3]: Kan Wu, **Kaiwei Tu**, Yuvraj Patel, Rathijit Sen, Kwanghyun Park, Andrea C. Arpaci-Dusseau, and Remzi H. Arpaci-Dusseau. "NyxCache: Flexible and Efficient Multi-tenant Persistent Memory Caching." *USENIX FAST '22*.

[4]: Kan Wu, Zhihan Guo, Guanzhou Hu, **Kaiwei Tu**, Ramnatthan Alagappan, Rathijit Sen, Kwanghyun Park, Andrea C. Arpaci-Dusseau, and Remzi H. Arpaci-Dusseau. "The Storage Hierarchy is Not a Hierarchy: Optimizing Caching on Modern Storage Devices with Orthus." *USENIX FAST '21*.

Teaching Experience

University of Wisconsin–Madison **Madison, WI**
Teaching Assistant, Introduction to Operating Systems Spring 2021

Skills

Programming: C, C++, Python, Rust, Java, SQL, Linux Kernel

ML Systems: PyTorch, SGLang

Honors & Awards

2022, 2026: USENIX Student Grant, FAST '22 and FAST '26

2019: James B. Angell Scholar

2018: EECS Scholars

2017–2019: Dean's List; University Honors

Academic Service

2026: ACM Conference on AI and Agentic Systems (CAIS 2026) Evaluation Committee

2025: EuroSys 2026 Shadow Program Committee